

ECSE-4040

Digital Electronics

Lecture 10

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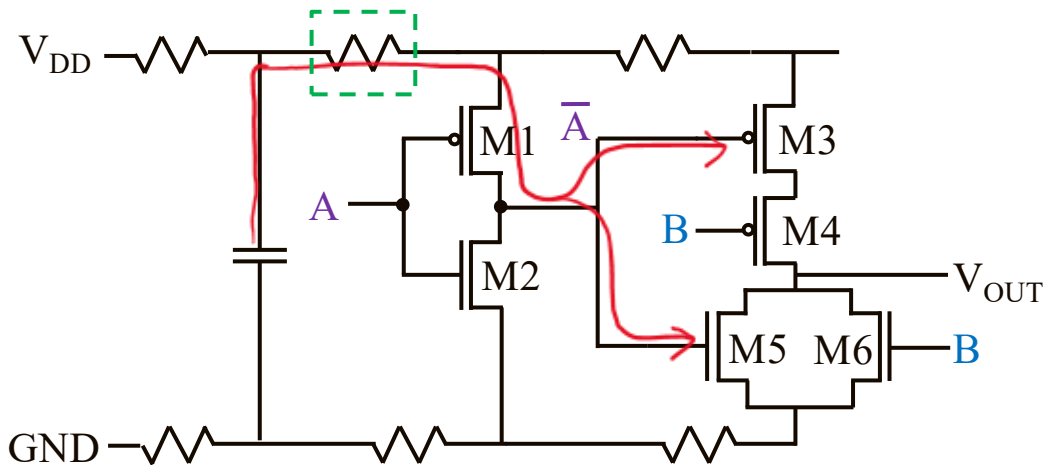
<https://lms.rpi.edu>

Lecture Outline

- Noise Sources
- Performance, Noise Margin Optimization
- Review on Combinational Logic

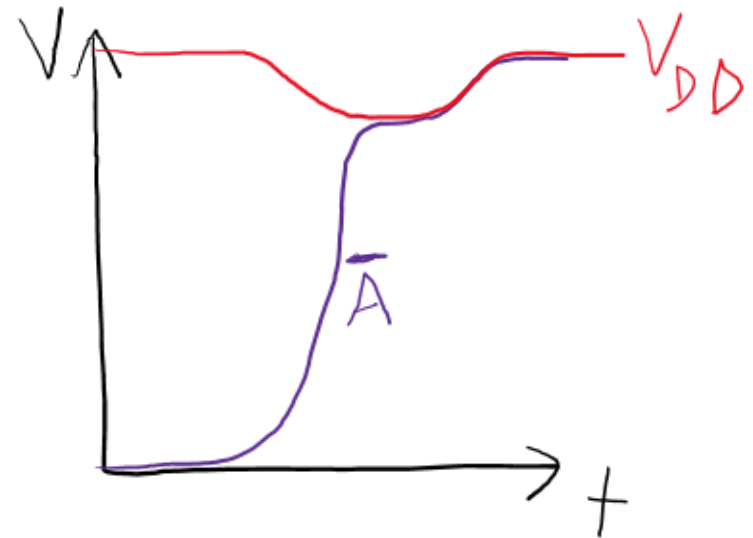
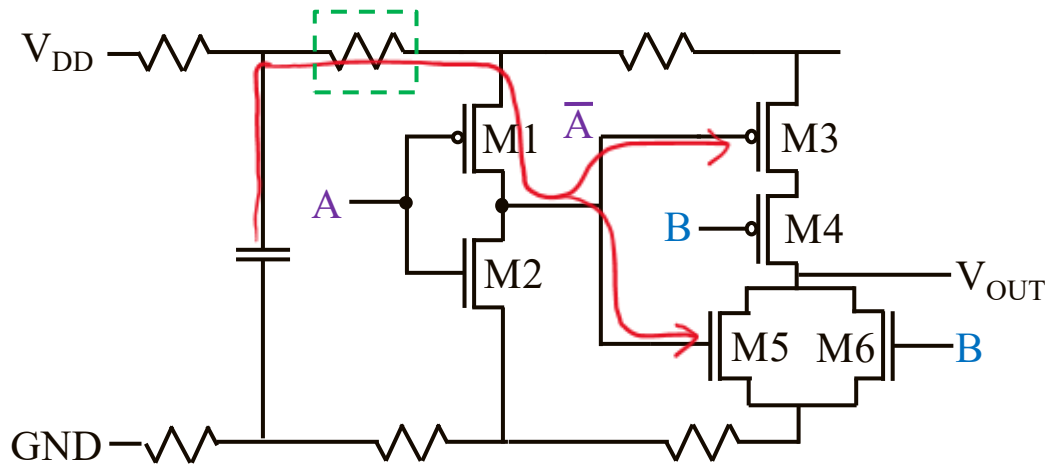
Noise Sources – Switching Noise

- We have discussed having large noise margins for immunity to noise.
- In CMOS circuits, circuits draw no static power (ideally).
- An output switching event requires a surge of current, which creates a voltage drop with non-ideal power and ground.
 - Reduces V_{SD} and V_{DS} , which reduces V_{OH} and V_{OL} respectively.



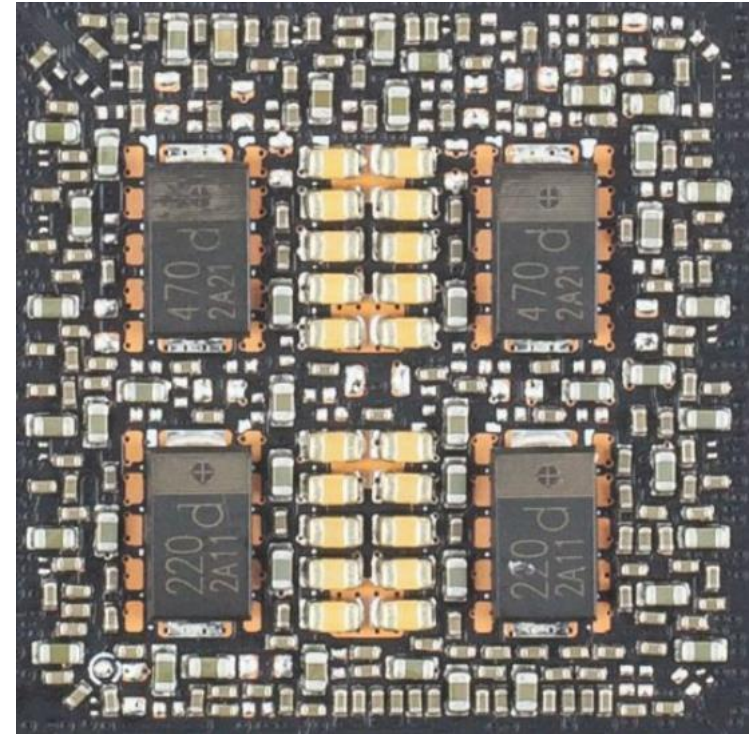
- INV and NOR gate
- In the case that $A, B = 0, 0$
- V_{DD} resistance decreases V_{Source} for M1, reducing V_{OH} and thus A' .
- The NOR must have a V_{IH} that is lower than the A' input
- NOR also sees a degraded V_{DD} !

Noise Sources – Switching Noise



Noise Sources – Switching Noise

- This are the V_{DD} , GND capacitors for the processor for an NVIDIA RTX 3080.
 - There are many, MANY capacitors to provide a stable V_{DD} .
 - Need many points of VDD,GND connection to reduce resistive losses.
 - Capacitors also must be responsive.
 - Some vendors used too few expensive capacitors
 - Resulted in degraded V_{OH}
 - ~\$2000 graphics card failed due to inadequate noise margins
 - This is why you do those V_{IH} , V_{IL} calculations!



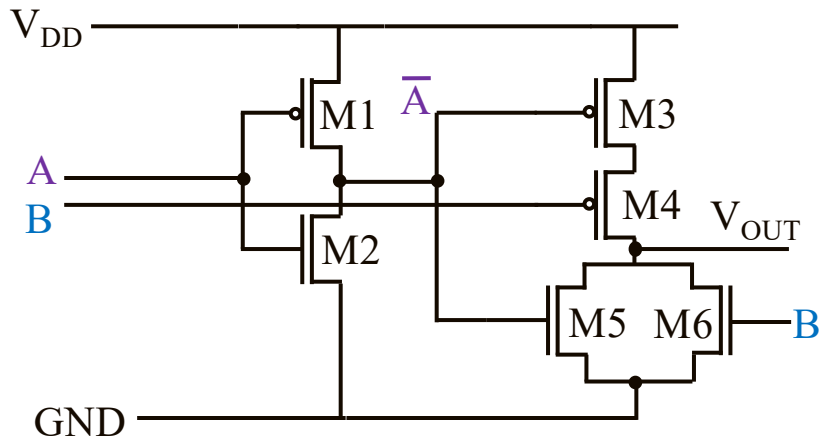
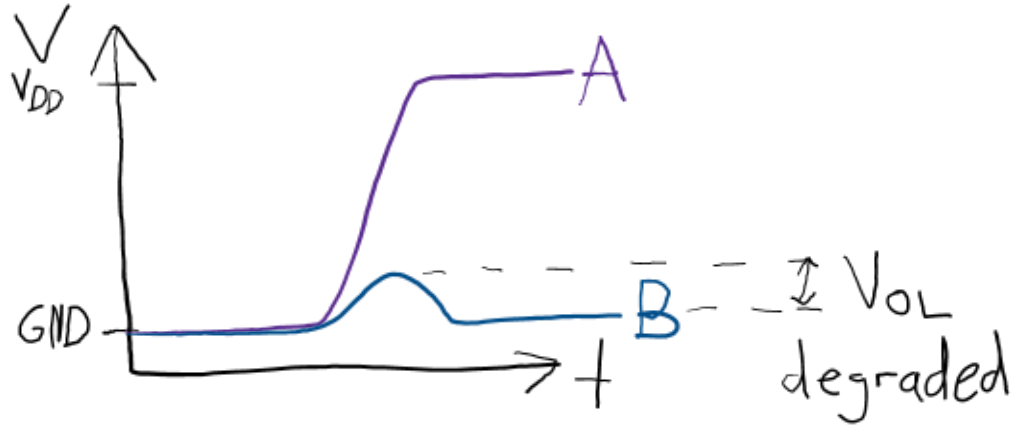
<https://www.hardwaretimes.com/asus-rtx-3080-gpus-use-the-best-caps-followed-by-fe-zotac-has-the-worst/>

Noise Sources - Crosstalk

- Signals do not just have capacitance to ground, but between each other as well.
 - This causes signals to couple together, mirroring each other's voltage
 - Ex if signals A and B are coupled with a capacitor:
 - When A $0 \rightarrow 1$:
 - If B is 0, voltage will increase slightly from GND, degrading V_{OL}
 - When A $1 \rightarrow 0$:
 - If B is 1, voltage will decrease slightly from V_{DD} , degrading V_{OH}

Noise Sources – Crosstalk

- If A and B run close together over a “long” distance, they will have a significant capacitance.



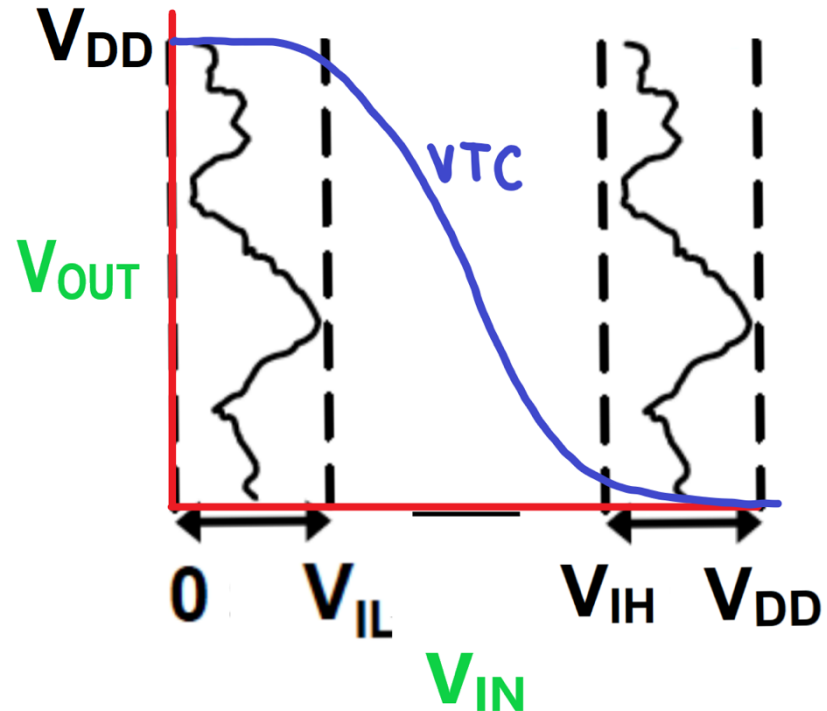
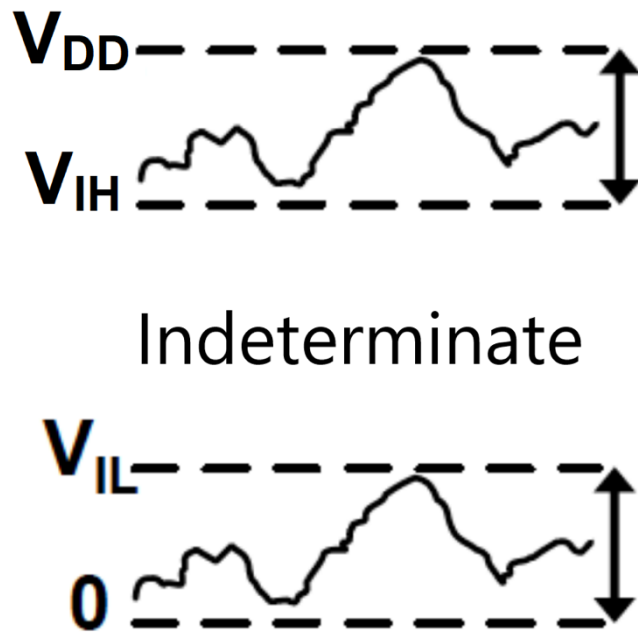
$$C_{\text{coupling}} = \epsilon * A/d, \text{ d is miniscule}$$

A

This distance between metals can be just 10's of nm!

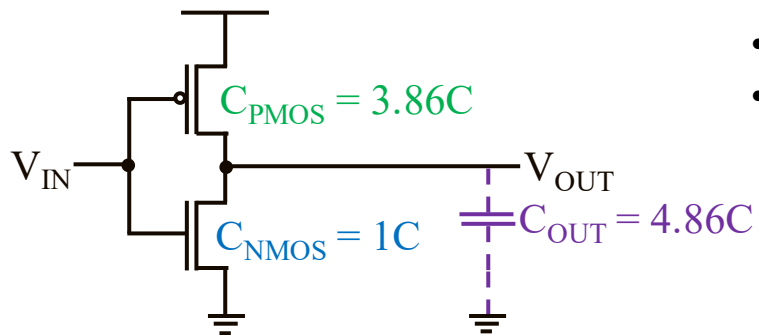
B

Noise Review



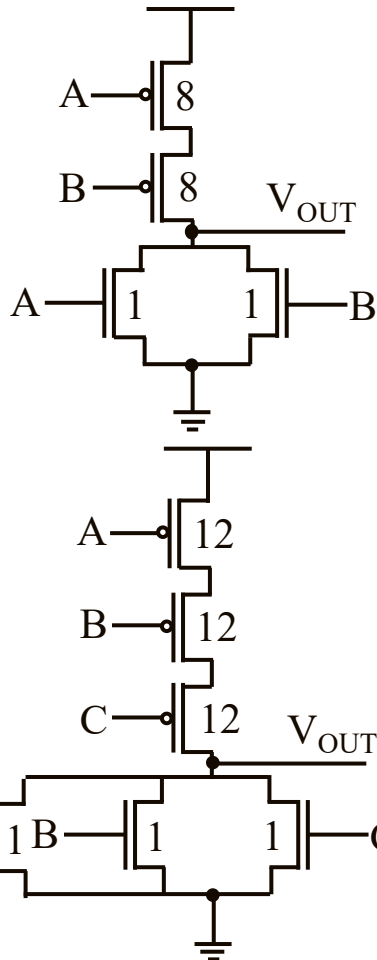
Normalized R, C

- Design methods are the same, regardless of process.
- Use relative, normalized R and C values to express delays.
- Node V_{OUT} always has a C_{OUT} from MOSFET output capacitance, and will have a load from downwind wire and MOSFET input capacitance.



- $R = R_{ON}$ of $W/L = 1$ NMOS transistor.
- $C = C$ of $W/L = 1$ transistor.
- In the case of a Beta balanced inverter ($V_S = V_{DD}/2$):
 - $R_{ONP} = R_{ONN} = 1$
 - $W_P = \mu_N/\mu_P = 270/70 = 3.86 * W_N$
 - $C_{OUT} = C_{NMOS} + C_{PMOS}$
 - $W_P = 3.86 * W_N$
 - $W_N = 1C$, and $W_P = 3.86C$
 - C_{OUT} of 4.86 units.
 - $RC = 4.86$

Normalized R,C Examples

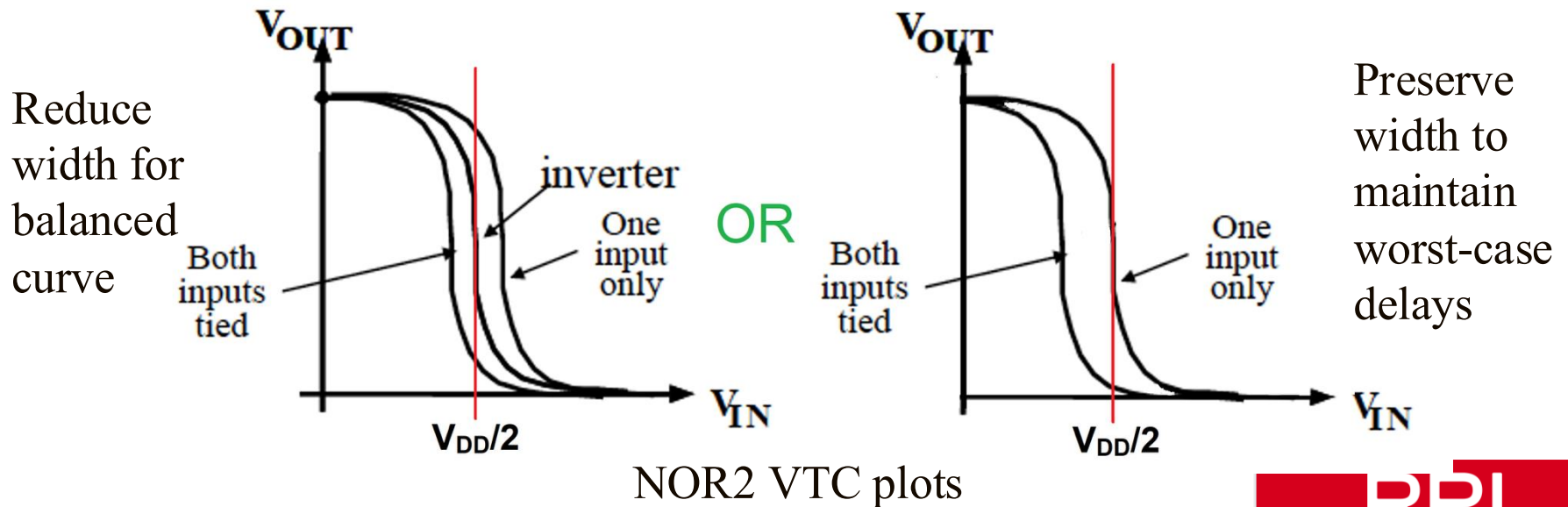


- NOR2
 - $R_{ONP} = 4 * (1+1) / 8 = 1$
 - $R_{ONN} = 1 \text{ OR } 1 / (1+1) = 0.5$
 - $C_{OUT} = 8 + (1+1) = 10$ (actually 9 in layout)
 - Intermediate PMOS node ignored for now
 - $\tau_{PLH} = \tau_{PHL} = 0.69 * RC$
 - With no external load considered:
 - $\tau_p = 0.69 * 1 * 10 = 6.9$ time units
 - $C \sim 1\text{fF}$, $R = 12.5\text{k}$
 - With large 1pF C_L added
 - $C_L \gg C$, transistor internal capacitance insignificant
 - $\tau_p = 0.69 * 12.5\text{k} * 1\text{pF} = 12.5\text{ns}$
- NOR3
 - $R_{ONP} = 4 * (1+1+1) / 12 = 1$
 - $R_{ONN} = 1 \text{ OR } 1 / (1+1) = 0.5 \text{ OR } 1 / (1+1+1) = 0.33$
 - $C_{OUT} = 12 + (1+1+1) = 15$
 - Now ignoring 2 Intermediate PMOS nodes
 - $\tau_p = 0.69 * 1 * 15 = 10.35$ time units (no external load)
 - $\tau_p = 0.69 * 12.5\text{k} * 1\text{pF} = 12.5\text{ns}$ (with large external load)
 - Standard practice – assume large C_L that dominates transistors capacitance, preserve $R_{ON} = 1$ when possible

- $\mu_N / \mu_P = 270 / 70 = 3.86$
 - $3.86 = 4$
- Transistors in series
- Transistor width
- Transistor in parallel
- Must design for worst-case R

Complex Circuit Noise Margins

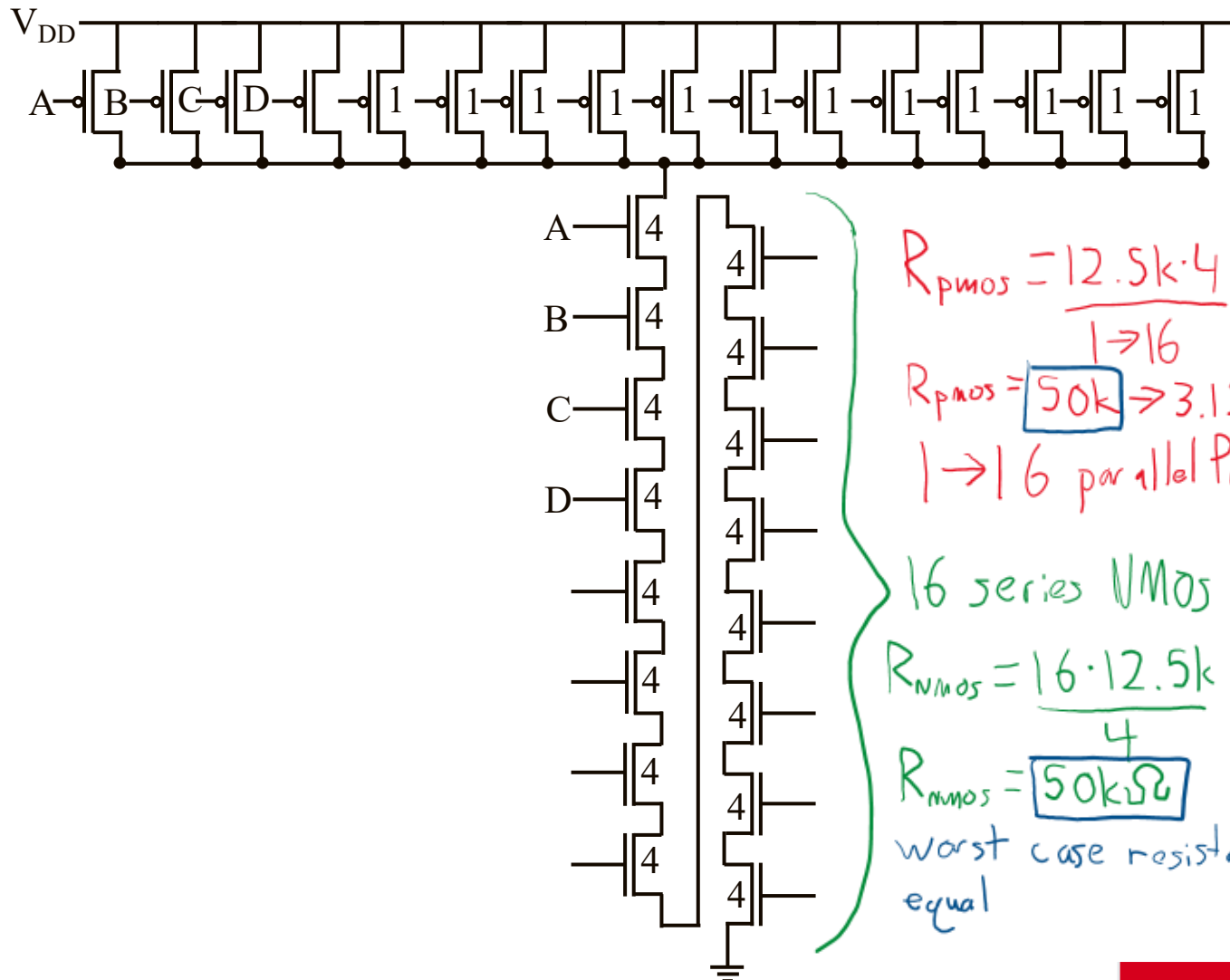
- Ideally, V_S is located in the middle of the VTC curve, with V_{IL} and V_{IH} values that closely surround V_S for maximum noise margins.
- For more complex gates, such as NAND and NOR, there are multiple input dependent VTC curves
 - One, two, or more parallel transistors active depending on the input.
 - How to decide between optimizing for speed, or noise margins?



Noise Margins Guidelines

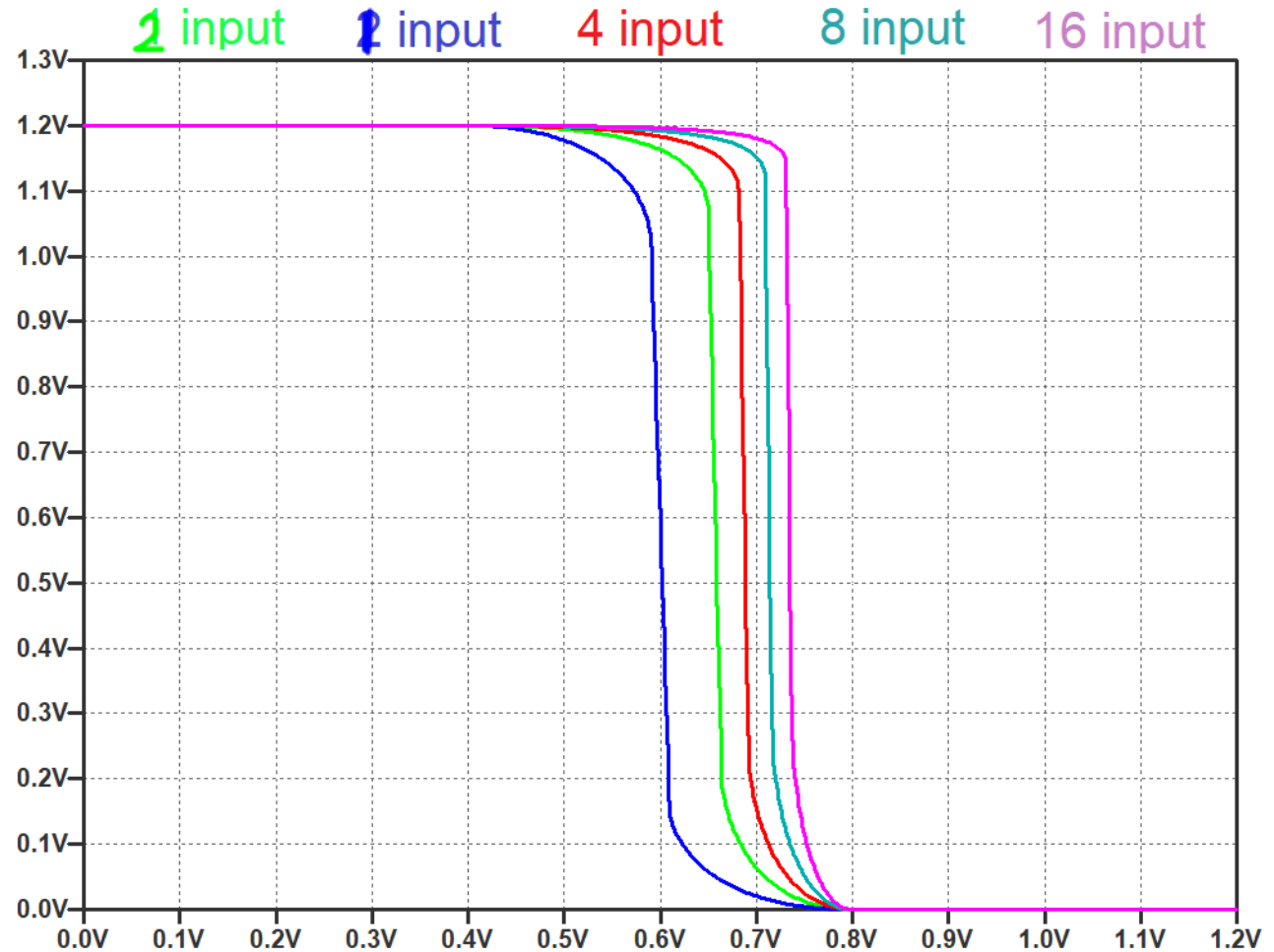
- Design practice – Preserve worst-case delay, unless noise margins become inadequate
 - Only allow imbalancing VTC curve to an extent
 - Noise margins are given as a design requirement
 - Exceeding noise margin requirement is wasteful
 - Not meeting it results in buggy circuits

NAND-16



Noise Margins, Multi-input NAND

- N_{MH} approaches $|V_{TP}|$ as more inputs are added.
- If $|V_{TP}|$ is an acceptable noise margin, never need to calculate noise margins.



Noise Margin Target

- What if $|V_{TP}|$ wasn't an acceptable noise margin?
 - How many inputs can we have until we need to adjust width for noise margins?
 - How do we adjust for noise margins?

$$V_{IH} = \frac{2V_{out} + V_{TN} + (k_P/k_N)(V_{DD} - |V_{TP}|)}{1 + (k_P/k_N)}$$

Set V_{IH}

Solve for k_P/k_N in terms of W_P/W_N

Solve for V_{out}

$$\frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_N}\right)} \left[(V_{in} - V_{TN})V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P}{L_P} \frac{v_{sat} C_{ox}}{(V_{DD} - V_{in} - |V_{TP}|) + E_{CP} L_P} (V_{DD} - V_{in} - |V_{TP}|)^2$$

V_{in} is V_{IH}

Plug in prior V_{out} calculation

solve for W_P/W_N

Noise Margin Target

$$N_{MH} = 0.5 = V_{OH} - V_{IH}$$

$$0.5 = 1.2 - V_{IH}$$

$$\underline{V_{IH} = 0.7V}$$

↓ NMH is from worst case with all PMOS on

$$\frac{k_p}{k_n} = \frac{16 \cdot 70 \cdot W_p}{270/16 \cdot W_n} = \underline{66.4 \cdot \frac{W_p}{W_n}}$$

↑ 16 NMOS in series

$$0.7 = \frac{2V_{out} + 0.4 + 66.4 \cdot \frac{W_p}{W_n} (1.2 - 0.4)}{1 + 66.4 \cdot \frac{W_p}{W_n}}$$

$$\frac{W_p}{W_n} = 0.452 - 0.3V_{out},$$

$$\underline{V_{out} = 0.15 - 3.32 \frac{W_p}{W_n}}$$

$$\underline{\frac{W_p}{W_n} = \frac{1}{4}}$$

when worst case $\beta(k)$ balanced.

Our result will be lower to increase N_{MH}

Noise Margin Target

$$\frac{W_N}{L_N} \cdot \frac{270}{\left(1 + \frac{V_{out}}{0.6}\right)} \left[(0.7 - 0.4)V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P \cdot 8 \cdot 10^6 \cdot (1.2 - 0.7 - 0.4)^2}{1.2 - 0.7 - 0.4 + 0.6}$$

$$\frac{W_P}{W_N} = \frac{236.25}{1 + \frac{V_{out}}{0.6}} \left(0.3V_{out} - \frac{V_{out}^2}{2} \right)$$

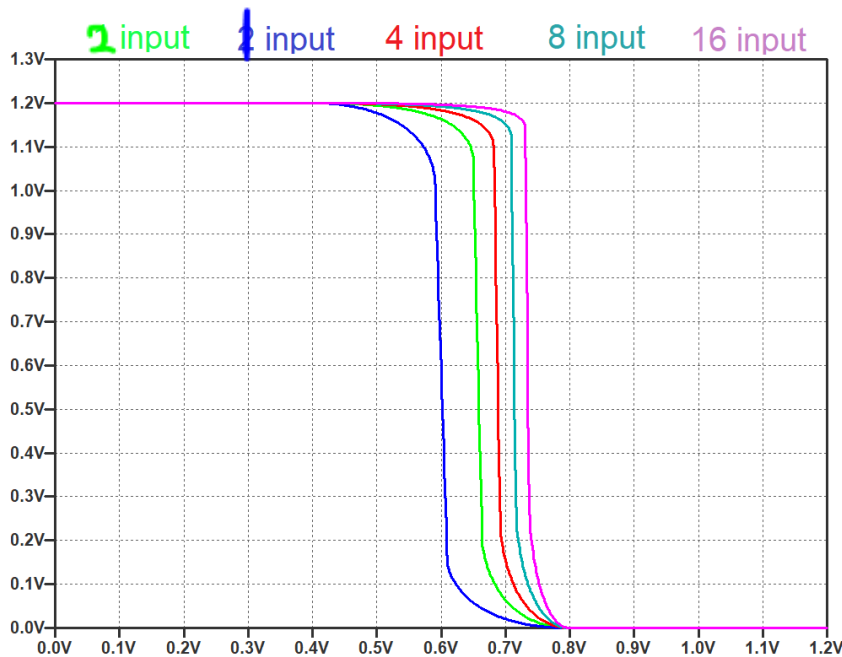
$$\frac{W_P}{W_N} = 0.045$$

$$W_P = 100n$$

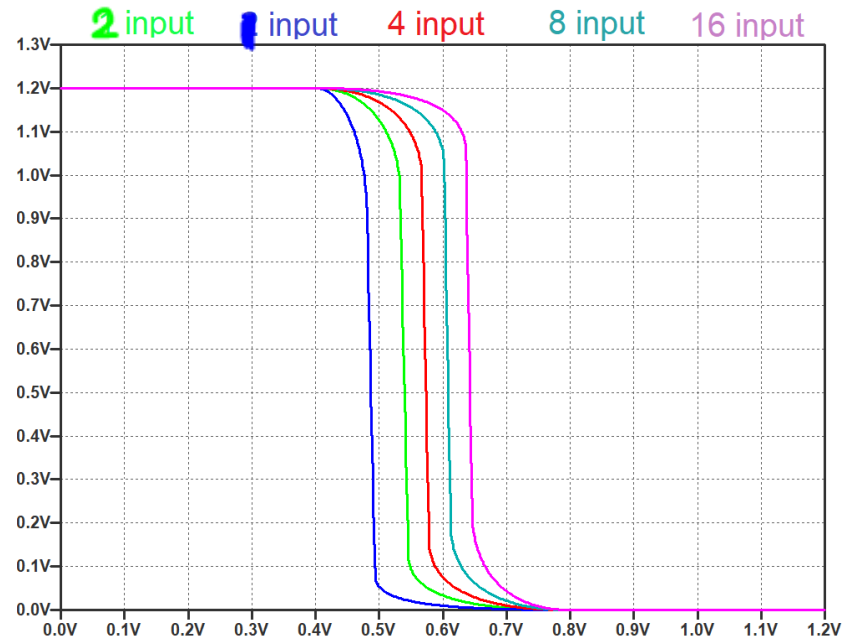
$$W_N = \frac{W_P}{0.045}$$

$$W_N = 2222n$$

Noise margin Target



- $R_{ONP} = R_{ONN}$ (worst-case)
- $W_p = (4 * W_n / 16) = (W_n / 4)$
- Cannot have $W_p < 100n$, so $W_p = 100n$, $W_n = 400n$
- $R_{ON} = R_{INV} * 4 = 12.5k * 4 = 50k \text{ Ohms}$



- $N_{MH} = 0.5V$ requirement, $0.5 = V_{OH} - V_{IH}$, $V_{IH} = 0.7V$
- $W_p = W_n / 32$
- $R_{ONP} = 4 * 32 * R_{ONN}$ (worst-case)
- $R_{ONN} = R_{INV} * 16 / 32 = 12.5k / 2 = 6.25k \text{ Ohms}$
- $R_{ONP} = R_{ONN} * 128 = 6.25k * 128 = 800k \text{ Ohms}$

Complex Logic Circuits-Review

- Ability to easily build complex logic gates is one of the most attractive features of MOS logic circuits
- Design principle of pull-down network
 - OR operations are performed by parallel connected drivers
 - AND operations are performed by series connected drivers
 - Inversion is provided by the nature of MOS circuit operation
- Don't get too carried away...Use this knowledge wisely
 - Remember that complex functions don't have to be implemented with a single gate
 - Can break up very complicated Boolean expressions into a cascade of gate stages
- We will use DeMorgan's Law to build dual networks

Complex CMOS Gate Design-Review

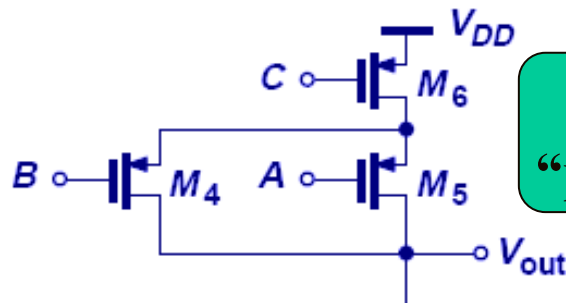
- Implement an AND-OR-INVERT (AOI) function

$$Z = \overline{(A \cdot B + C)}$$

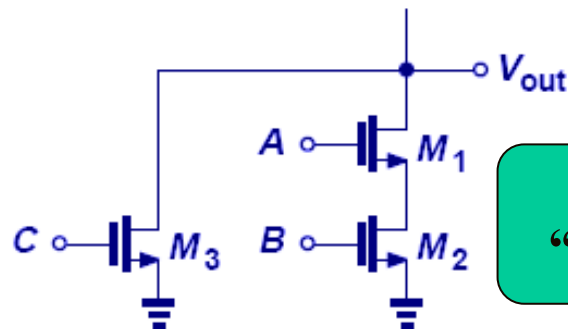
- Get the expression into forms that enable easy implementation of pull-up and pull-down networks

$$Z = \overline{(A \cdot B + C)}$$

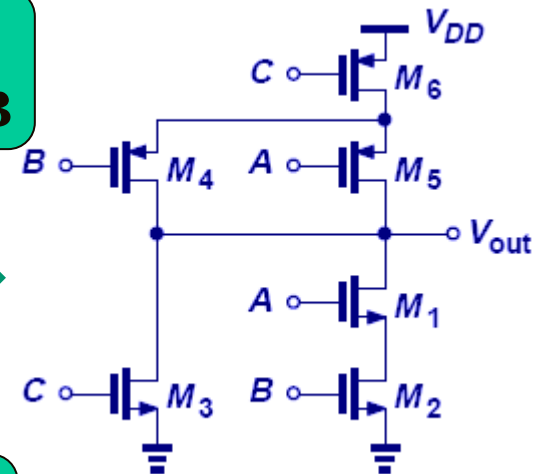
$$Z = (\overline{A} + \overline{B}) \cdot \overline{C}$$



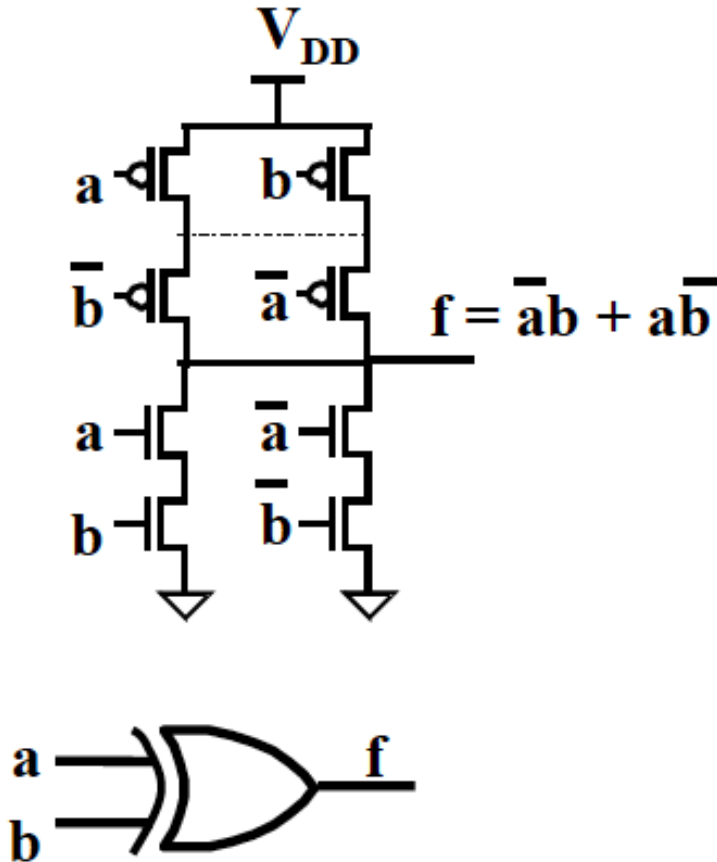
C is in “series” with the “parallel” combination of A and B



C is in “parallel” with the “series” combination of A and B



XOR Gates - Review



$$f = \overline{a \cdot b} + \overline{a \cdot \bar{b}}$$

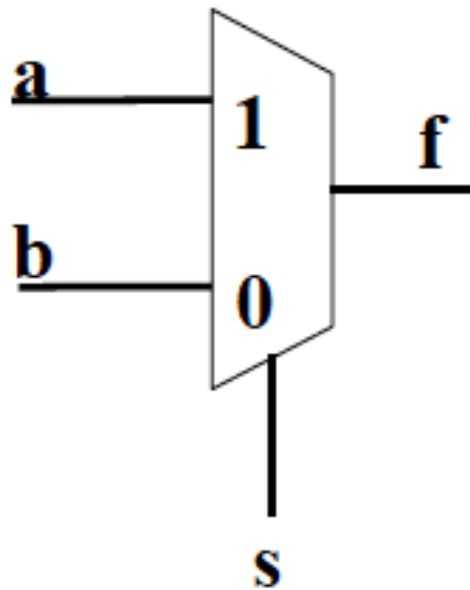
$$f = (a + \bar{b}) \cdot (\bar{a} + b)$$

$$f = \overline{a \cdot \bar{a} + a \cdot b + \bar{a} \cdot \bar{b} + \bar{b} \cdot b}$$

$$f = \cancel{a \cdot \bar{a}} + a \cdot b + \bar{a} \cdot \bar{b} + \cancel{\bar{b} \cdot b}$$

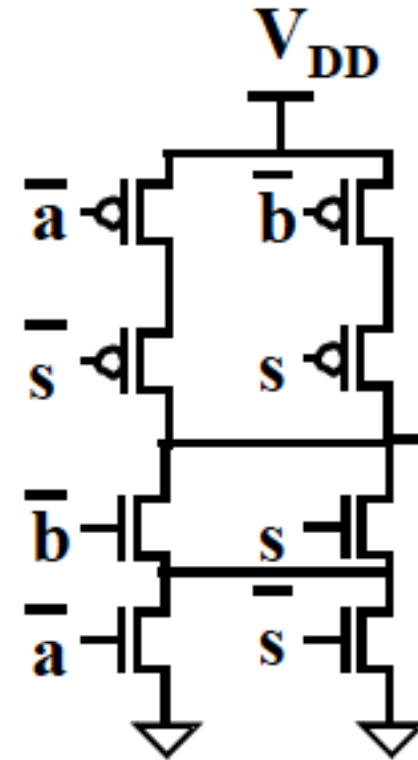
$$f = a \cdot b + \bar{a} \cdot \bar{b}$$

CMOS Multiplexer



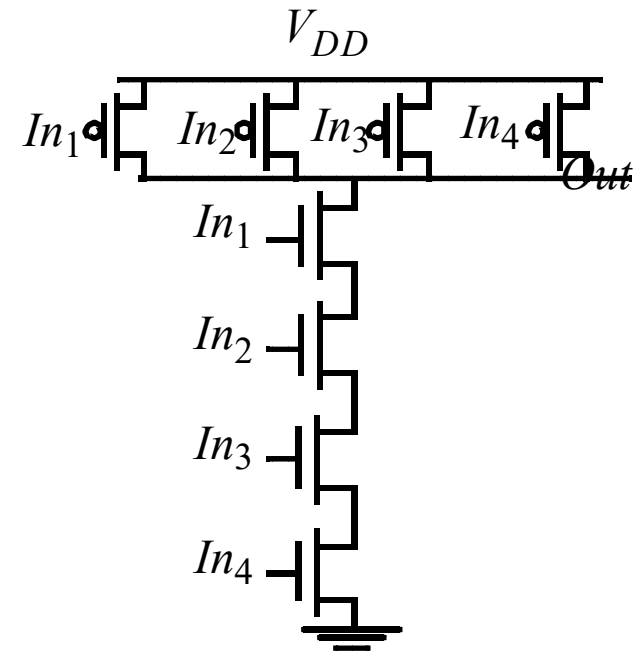
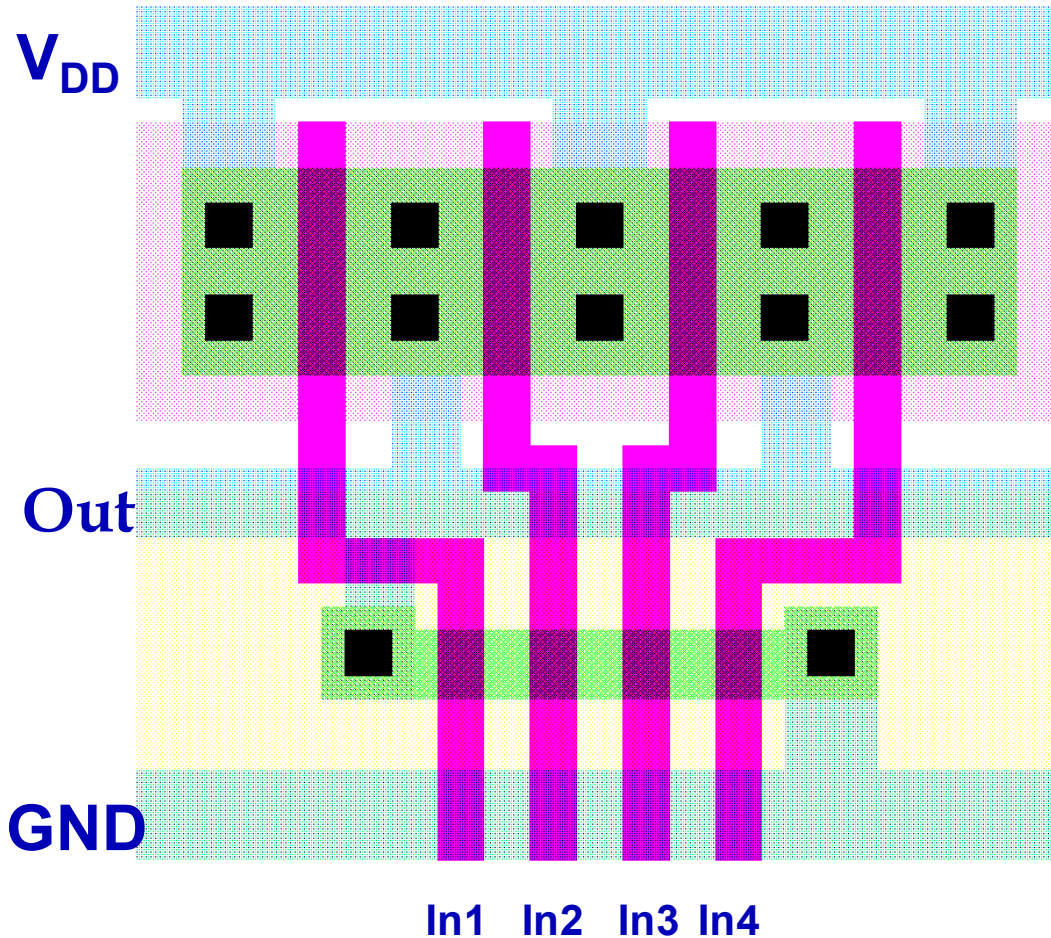
$$f = a.s + b.\overline{s}$$

$$f = \overline{\overline{a.s + b.\overline{s}}} \longrightarrow f = \overline{(\overline{a + s}).(\overline{b + s})}$$



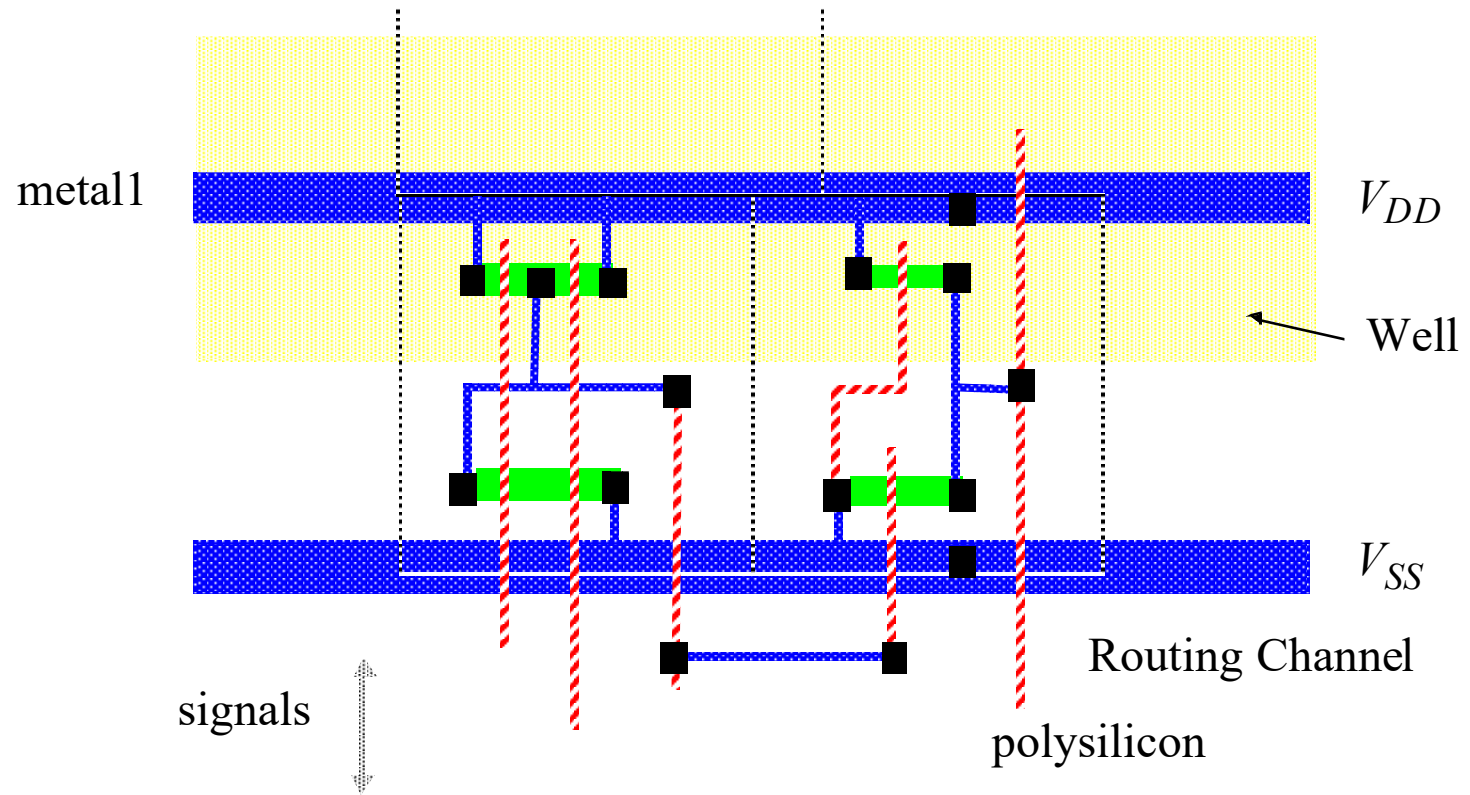
Examples of layouts of CMOS

Complex gates

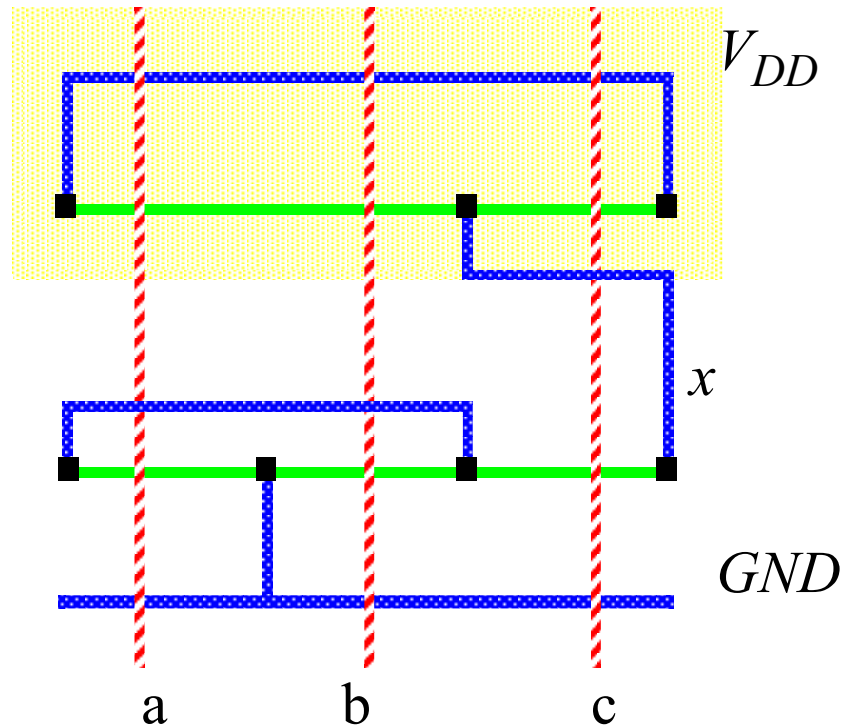


4-input NAND Gate

Examples of Standard Cell Layout Methodology



Examples of Standard Cell Layout



$$f = \overline{(a+b)} \cdot c$$

The Good & Bad News

❖ High Noise margins:

❖ V_{OH} and V_{OL} are at V_{DD} and GND

❖ No Static Power dissipation

❖ There is no direct path between V_{DD} and GND in steady state mode

❖ Comparable rise and fall times

❖ (under appropriate scaling conditions)

- Slows down dramatically for large fan-ins due to long series stack of transistors
 - Fan-ins = number of inputs
 - PMOS stacks are worse than NMOS series stacks
- Large number of transistors
 - $2n$ devices for n -input NAND
 - At least 2 devices per input
- Bigger layout
 - N+ and P+ spacing rule and well spacing rule
 - Large device sizes required to counteract series stack
- Limit fan-in to 3 or 4...or delay and area will be too large